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Decreasing security threshold against double spend attack in networks with slow synchronization

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ABSTRACT

We research the probability change of successful double spend attack on Proof-of-Work consensus protocol depending on network parameters in the model with continuous time. We analyze influence of block generation rate in networks with delayed message delivery on probability of double spend attack success, and provide strict analytical expressions for network security threshold and for the upper bound of block generation rate.

1. Introduction

We consider the general model of a blockchain network with Nakamoto consensus with one common property: a miner who creates the next block is chosen randomly from the set of all miners during the block generation process and the probability for miner to create a new block depends on his computational power. The example of such consensus protocol is the Proof-of-Work protocol. After creating a block, the miner should share it among other participants of the network. This sharing process may take some non-zero time which depends on the network parameters.

At the same time, a situation may occur when malicious miners (adversaries) are consolidated into a large enough mining pool, which consists of multiple devices on the same location, and, as a result, these devices are highly synchronized. Their synchronization time, compared to honest miners' time, is close to zero. Thus the adversaries' ratio that guarantees their success in a double spend attack may be essentially less than the ratio of honest miners, i.e., it may be less than 50%.

The reason is that if time delay is equal to Δ seconds, then the next block generated by honest miners cannot occur earlier than in Δ seconds, while adversary has zero time delay and may generate his next block any time with non-zero probability.

In this paper, we introduce the concept of threshold, that is the minimal adversaries' ratio which provides a successful attack with probability 1. We obtained a strictly proved analytical expression for this threshold that demonstrates its dependence not only on the adversary's ratio, but also on such network parameters as time delays and intensity of block generation. Here time delay is time of network synchronization, and intensity of block generation is the number of blocks generated per time unit (e.g., second). The expression obtained allows to get the relevant numerical results and to find the adversary ratio that is critical for a given network. On the other hand, it also allows defining the maximal intensity of block generation permitted for the given network without its complete vulnerability to the attack.

In this paper we also show that the time delay value affects the secure threshold and probability of attack. In other words, the larger time delay for honest miners, the smaller threshold and the larger probability of double spend attack. We will strictly prove both of these statements in Sections 4 and 6.

2. Related work

For the first time, a double spend attack was described in the well-known paper by Nakamoto [1], where he also obtained expressions for attack probability (which were not entirely accurate). Later, more consistent results on the attack probability were obtained in other papers [2–4], and the paper [4] gives completely proved expressions for probability of such attack. However, the main drawback of these papers is that the results were obtained under the assumption on the time delay (i.e., the time between block creation and its sharing among all miners) to be equal to zero, that is not so for a real network.

At the same time, some papers (e.g., [5,6]) contain an observation that the probability of successful attack significantly depends on the time delay and intensity of block generation. In [7,8] a problem of constructing asymptotic estimates of splitting attack probability for the model with non-zero synchronization time (equal for honest miners and the adversary) is considered with time being a discrete parameter.

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In [9] statements on asymptotic properties of the blockchain with limited block delivery time were obtained. However, at the moment, there is no published method to obtain exact values for the double spend attack probability depending on network parameters, in particular, on the synchronization time.

In our paper [10] we succeeded to provide a partial answer to this question, having considered two special cases: when the synchronization time interval of the adversary is the same as that of honest miners, and when it is two times smaller. Note that these results were obtained for the model with discrete time, and that significantly simplifies the process of obtaining the results.

3. Our results

In this paper, we give an answer to the next two questions:

1. how synchronization time affects on the security threshold and probability of double spend attack;
2. what block generation intensity is maximal for given network parameters that network is still secure.

The model, in which our results were obtained, has the following characteristics:

- time is a continuous parameter;
- synchronization time between honest miners is upper bounded by a given arbitrary value; moreover, the adversary can delay the block delivery for honest miners within this upper bound;
- synchronization time of the adversary is also a given arbitrary value (lower than for honest participants) and can be set to zero;
- block generation rate is set to an arbitrary value (both for honest miners and for the adversary);
- the share of adversarial hashpower is arbitrary.

We give strictly proved expressions for the minimal share of adversaries in the presence of which their attack is guaranteed to be successful. At the same time, depending on the network parameters, this ratio may be essentially smaller than 50%.

Using the results obtained, we also give a solution of another problem of current interest related to increase in block generation intensity.

We find, under certain network parameters, the limit up to which the block generation intensity can be increased so that the network is not completely vulnerable to attacks.

Our results generalize the corresponding results obtained in [2,4] and some others paper. For example, if in our model we set synchronization time to be equal to 0, we get particular case described in [4]. Our model is much more general then the one given in these works and more practical, because the synchronization time in large decentralized networks cannot be equal to zero for honest participants.

One of the most essential results of our paper is that we showed that the concept of "50 percent attack" does not work in real world networks. We showed that if synchronization time is non-zero, the adversary may require essentially less than 50 percent of hashrate to provide attack with probability 1. Our results allows to determine, for fixed network parameters, such minimal hashrate for which network is vulnerable to double spending attack.

For all analytical results we also provide numerical examples and relevant graphics.

4. Security threshold: Definition, analytic expressions and numerical results

Let us recall that, according to our assumptions, the consensus protocol provides unpredictable block generation (i.e., the creator of the next block is unknown to all miners until the block is created).

In this work we also use the following designations (see Table 1).

Table 1
Main designations.

α_H	Intensity of block generation by honest miners
α_M	Intensity of block generation by the adversary
$\alpha = \alpha_H + \alpha_M$	General intensity of block generation in network, i.e., the number of blocks generated per time unit
D_H	Time delay for honest miners
D_M	Time delay for the adversary
$\Delta = D_H - D_M \geq 0$	Difference between time delays
$p_H = \frac{\alpha_H}{\alpha}$	Ratio of honest miners
$p_M = \frac{\alpha_M}{\alpha}$	Ratio of the adversary

We make an assumption in favor of the adversary, and assume that $D_M \leq D_H$ (with the reverse inequality, the results can be obtained similarly, but in that case honest miners will have an advantage (that is the unrealistic assumption), and the adversary will need to control more than 50% to attack).

In the case of the PoW consensus protocol, p_H and p_M values are equal to corresponding hashrates; for other protocols, the values are the ratio of some other resources bind to block generation rate.

In a particular case, when $D_H = D_M = 0$, the probability of a double spend attack is equal to 1 if and only if $p_M \geq 1/2$ (i.e., the adversarial ratio is not less than 50%). Hence the term "50% attack" means the attack which is guaranteed to be successful. It implies that $p_M = 1/2$ is the security threshold in this case.

However, for the general case the security threshold is to be determined in a more complicated way, taking into account the network parameters.

Definition 1. For a given network with parameters α, α_H, α_M, D_H and D_M its *security threshold* p_{st} is the minimal adversarial ratio that guarantees success of a double spend attack (i.e., if the adversarial ratio is not less than p_{st}, then the probability of the successful attack is equal to 1).

Further we will obtain the exact value of the security threshold for the preset network parameters. For this purpose, we need to describe the following values and relations between them:

- T_H is a random variable that is equal to time spent by honest miners to create one block;
- T'_H is a random variable that is equal to time spent by honest miners to create one block and propagate it across all honest nodes in the network;
- T_M is a random variable that is equal to time spent by malicious miners to create one block;
- T'_M is a random variable that is equal to time spent by malicious miners to create one block and propagate it across all malicious nodes in the network.

Note that we make an assumption in favor to adversaries and assume that they obtain all blocks, even generated by honest miners, within time interval D_M.

As it was shown in [4], random values T_H and T_M have exponential distributions:

$$F_{T_H}(t) = P(T_H < t) = 1 - e^{\alpha_H t},$$
$$F_{T_M}(t) = P(T_M < t) = 1 - e^{\alpha_M t}, \quad (1)$$

for the parameters $\alpha_H > 0$, $\alpha_M > 0$.

In our notations, the following equalities hold:

$$T'_H = D_H + T_H,\ T'_M = D_M + T_M. \quad (2)$$

Let us denote by p_H the probability that honest miners create the next block before malicious miners do that(provided that they started

creating this block simultaneously), and $p_M = 1 - p_H$ is the probability of the opposite event. According to [4],

$$p_H = \frac{\alpha_H}{\alpha_H + \alpha_M},\ p_M = \frac{\alpha_M}{\alpha_H + \alpha_M}. \tag{3}$$

Let us also denote as p'_H the probability that honest miners create the next block and propagate it across all nodes in the network (at least, across all honest nodes) before malicious miners do that, and as p'_M-the probability of the opposite event, $p'_M = 1 - p'_H$.

Then

$$p'_H = P(T'_H < T'_M),\ p'_M = P(T'_M < T'_H), \tag{4}$$

and $p'_H + p'_M = 1$.

Let us denote as $T'_H(i)$ time which honest miners need to create and propagate the ith block, i.e., time between events "$(i-1)$th block is created and received by all honest participants" and "ith block is created and received by all honest participants". Similarly to (2), we also can write the equality

$$T'_H(i) = T_H(i) + D_H. \tag{5}$$

Then $T'_H(i)$, $i \geq 1$ are independent identically distributed random variables with distribution functions

$$F_{T'_{H(i)}}(t) = F_{T'_H}(t) = F_{T_H}(t - D_H) = 1 - e^{-\alpha_H(t-D_H)},$$
$$\text{for all } i \geq 1,\ t \geq D_H,$$

where the latter equality follows from (1).

Similarly, we define the random variables $T_M(i)$, $i \geq 1$, with distribution functions

$$F_{T_{M(i)}}(t) = 1 - e^{-\alpha_M t},\ \text{for all } i \geq 1.$$

Also, for $n \geq 1$ let us define random variables $S_H(n)$, where

$$S_H(n) = \sum_{i=1}^{n} T_H(i), \tag{6}$$

and random variables $S'_H(n)$, where

$$S'_H(n) = \sum_{i=1}^{n} T'_H(i). \tag{7}$$

Then $S_H(n)$ is the time needed for generation (on condition of zero synchronization time) of n independent blocks by honest miners, and $S'_H(n)$ is the time that is needed for honest miners to create and propagate consistently n blocks across the network.

From (5) we obtain:

$$S'_H(n) = S_H(n) + nD_H,$$

where $S_H(n)$ has the Erlang distribution (as a sum of identically distributed random variables with exponential distribution, [11,12]):

$$F_{S_H(n)}(t) = P(S_H(n) \leq t) = 1 - e^{-\alpha_H t} \sum_{i=1}^{n} \frac{(\alpha_H t)^k}{k!}. \tag{8}$$

Similarly, let us define random variables $S_M(n)$ and $S'_M(n)$:

$$S_M(n) = \sum_{i=1}^{n} T_M(i),\ S'_M(n) = \sum_{i=1}^{n} T'_M(i) = S_M(n) + nD_M. \tag{9}$$

A random variable $S_M(n)$ is also a sum of independent exponentially distributed random variables, so it also has the Erlang distribution [11, 12]:

$$F_{S_M(n)}(t) = P(S_M(n) \leq t) = 1 - e^{-\alpha_M t} \sum_{i=1}^{n} \frac{(\alpha_M t)^k}{k!}. \tag{10}$$

For a fixed $t \geq 0$, let us define a random variable $N'_M(t)$ as a number of blocks that were created and propagated across the network by the adversary during time t.

Now we can formulate and prove two auxiliary lemmas needed to obtain the main result.

Lemma 1. *For the given network with parameters α, α_H, α_M, D_H and D_M the probability p'_M that the next block will be created by the adversary is equal to*

$$p'_M = 1 - e^{-\alpha_M \Delta} p_H;$$

the probability p'_H that the next block will be created by honest miners is equal to

$$p'_H = e^{-\alpha_M \Delta} p_H.$$

Proof. According to (2) and (4), the distribution functions of random variables T'_H and T'_M are as follows:

$$F_{T'_H}(t) = P(T'_H < t) = P(T_H + D_H < t) =$$
$$= P(T_H < t - D_H) = \begin{cases} 1 - e^{-\alpha_H(t-D_H)}, & \text{if } t > D_H, \\ 0, & \text{otherwise;} \end{cases}$$
$$F_{T'_M}(t) = P(T'_M < t) = P(T_M + D_M < t) =$$
$$= P(T_M < t - D_M) = \begin{cases} 1 - e^{-\alpha_M(t-D_M)}, & \text{if } t > D_M, \\ 0, & \text{otherwise.} \end{cases}$$

Respectively, densities of these distributions are the following functions:

$$f_{T'_H}(t) = \alpha_H e^{-\alpha_H(t-D_H)};\ f_{T'_M}(t) = \alpha_M e^{-\alpha_M(t-D_M)}. \tag{11}$$

The second equality in (4) can be rewritten as:

$$p'_M = P(T'_M < T'_H) = P(T_M + D_M < T_H + D_H) =$$
$$= P(T_M < T_H + D_H - D_M).$$

Let us define a random variable ζ, $\zeta = T_H + \Delta$.

Then

$$F_\zeta(x) = F_{T_H}(t - \Delta) = \begin{cases} 1 - e^{-\alpha_H(t-\Delta)}, & \text{if } t \geq \Delta; \\ 0, & \text{otherwise;} \end{cases}$$

$$f_\zeta(t) = \begin{cases} \alpha_H \cdot e^{-\alpha_H(t-\Delta)}, & \text{if } t \geq \Delta; \\ 0, & \text{otherwise.} \end{cases}$$

Hence $f_\zeta(t) = f_{T_H}(t - \Delta)$.

Taking into account these notations and using the compound probability formula, we obtain:

$$p'_M = P(T_M < T_H + \Delta) = P(T_M < T_H + \Delta / T_M < \Delta) P(T_M < \Delta) +$$
$$+ P(T_M \langle T_H + \Delta / T_M \rangle \Delta) P(T_M > \Delta). \tag{12}$$

However,

$$P(T_M < T_H + \Delta / T_M < \Delta) = 1,\ P(T_M < \Delta) = 1 - e^{-\alpha_M \Delta},\ \text{and}$$
$$P(T_M \langle T_H + \Delta / T_M \rangle \Delta) P(T_M > \Delta) = P(\Delta < T_M < T_H + \Delta).$$

Let us compute the latter probability:

$$P(\Delta < T_M < T_H + \Delta) = \int_{x,y:\,\Delta<x<y} f_{T_M}(x) f_\zeta(y) dx dy =$$
$$= \int_\Delta^\infty \left(\int_\Delta^y f_{T_M}(x) dx \right) f_\zeta(y) dy = \int_\Delta^\infty (F_{T_M}(y) - F_{T_M}(\Delta)) f_{T_H}(y - \Delta) dy =$$
$$= \int_\Delta^\infty \left(1 - e^{-\alpha_M y} - \left(1 - e^{-\alpha_M \Delta} \right) \right) \alpha_H e^{-\alpha_H(y-\Delta)} dy =$$
$$= \int_\Delta^\infty \left(e^{-\alpha_M \Delta} - e^{-\alpha_M y} \right) \alpha_H e^{-\alpha_H(y-\Delta)} dy =$$
$$= \alpha_H e^{-\alpha_M \Delta} \int_\Delta^\infty \left(1 - e^{-\alpha_M(y-\Delta)} \right) e^{-\alpha_H(y-\Delta)} dy =$$
$$= \alpha_H e^{-\alpha_M \Delta} \int_0^\infty \left(1 - e^{-\alpha_M z} \right) e^{-\alpha_H z} dz,$$

where $z = y - \Delta$.

After integration we obtain:

$$P(\Delta < T_M < T'_H) = e^{-\alpha_M \Delta} \cdot \frac{\alpha_M}{\alpha_H + \alpha_M},$$

and substituting the resulting expression into the formula (9), we obtain:

$$p'_M = 1 - e^{-\alpha_M \Delta} + e^{-\alpha_M \Delta} \cdot \frac{\alpha_M}{\alpha_H + \alpha_M} =$$
$$= 1 - e^{-\alpha_M \Delta} \cdot (1 - \frac{\alpha_M}{\alpha_H + \alpha_M}) = 1 - e^{-\alpha_M \Delta} \cdot \frac{\alpha_H}{\alpha_H + \alpha_M} = 1 - e^{-\alpha_M \Delta} \cdot p_H.$$

So, $p'_H = 1 - p'_M = e^{-\alpha_M \Delta} \cdot p_H$. □

Below we formulate Lemma 2, which is very similar to the result presented in [2]. For completeness we give proof of lemma, because there is no such a proof in [2,4].

Lemma 2. *Let, at some point in time t_0, the branch created by the adversary be n blocks shorter than the branch created by honest miners. Denote as E_n the event that at some point in time $t > t_0$ an adversary was able to create a longer chain, and let $q_n = P(E_n)$. Then*

$$q_n = \begin{cases} 1, & \text{if } p'_M \geq p'_H \text{ or } n < 0; \\ \left(\frac{p'_M}{p'_H}\right)^n, & \text{otherwise.} \end{cases} \tag{13}$$

Proof. Firstly, note that if $n < 0$ than adversary has just built chain with longer length and the attack is successful.

According to the compound probability formula:

$$q_n = P(E_n) = P(E_n/T'_H > T'_M)P(T'_H > T'_M) + \\ + P(E_n/T'_H < T'_M)P(T'_H < T'_M) = \\ = P(E_{n-1})p'_M + P(E_{n-1})p'_H,$$

where the latter equality was obtained using Theorem 1.

So,

$$q_n = q_{n-1}p'_M + q_{n-1}p'_H. \tag{14}$$

To solve Eq. (14), we write the corresponding difference equation:

$$\lambda^2 p'_H - \lambda + p'_M = 0,$$

whose roots are $\lambda_1 = 1$ and $\lambda_2 = \frac{p'_M}{p'_H}$. Then the general solution of (14) is

$$q_n = a\lambda_1^n + b\lambda_2^n = a + b\left(\frac{p'_M}{p'_H}\right)^n.$$

If $p'_M > p'_H$, then $\frac{p'_M}{p'_H} > 1$. But, according to the definition of q_n, the condition $0 \leq q_n \leq 1$ holds. So, in this case $b = 0$, $a = 1$, and the only solution of (14) is $q_n = 1$.

If $p'_M = p'_H = 1/2$, then we obtain the equation $\lambda^2 - 2\lambda + 1 = 0$, whence $\lambda_1 = \lambda_2 = 1$ and $q_n = 1$ for all $n \geq 1$, with account of the initial condition $q_0 = 1$.

Finally, if $p'_M < p'_H$, then from the boundary conditions $q_0 = 1$, $q_\infty = 0$ we obtain $a = 0$, $b = 1$, and $q_n = \left(\frac{p'_M}{p'_H}\right)^n$. □

Now we can prove the main result of this section.

Let us denote as $\gamma = \gamma(\alpha, \Delta)$ the average number of blocks generated by all miners during the time Δ:

$$\gamma = \gamma(\alpha, \Delta) = \alpha \cdot \Delta.$$

Theorem 1. *For a given network with the parameter γ, the security threshold p_{st} is the solution of the equation*

$$1 - p_{st} = \frac{e^{\gamma \cdot p_{st}}}{2}. \tag{15}$$

Proof. Let B be a block on which the double spend attack is performed; z is the number of confirmation blocks after the block B. Let us define the event $A_z(k)$ in the following way:

$$A_z(k) = \{N_M(S'_H(z) = k)\} = \{X'_M(z) = k\},\ k \geq 0,$$

where $X'_M(n) = N_M(S'_H(n))$.

In other words, the event $A_z(k)$ is as follows: "the adversary created and propagated exactly k blocks, while honest miners created and propagated z confirmation blocks".

Also, let us define the event A_z as "the adversary was able to build a longer chain after the block B received z confirmation blocks". Then

$$A_z = \left\{\bigcup_{k \geq z} A_z(k)\right\} \cup \left\{\bigcup_{k=0}^{z-1}\Big(A_z(k) \cap E_{n-k}\Big)\right\}, \tag{16}$$

where E_{z-k} is as defined in Lemma 1.

Note that the events $\{\bigcup_{k \geq z} A_z(k)\}$ and $\{\bigcup_{k=0}^{z-1}(A_z(k) \cap E_{n-k})\}$ do not intersect, and the events $A_z(k)$ and E_{n-k} are independent. Besides, according to Lemma 1:

$$P(E_{z-k}) = \begin{cases} 1, & \text{if } p'_M \geq p'_H; \\ \frac{p'_M}{p'_H}, & \text{otherwise.} \end{cases} \tag{17}$$

So, taking into account (16), we obtain:

$$P(A_z) = \sum_{k=z}^{\infty} P\big(A_z(k)\big) + \sum_{k=0}^{z-1} P\big(A_z(k)\big)P\big(E_{z-k}\big).$$

From the latter equality, taking into account (17), and that $\sum_{k=0}^{\infty} P\big(A_z(k)\big) = 1$, we obtain:

$$P(A_z) = \begin{cases} \sum_{k=z}^{\infty} P\big(A_z(k)\big) + \\ + \sum_{k=0}^{z-1} P\big(A_z(k)\big)\left(\frac{p'_M}{p'_H}\right)^{z-k}, & \text{if } p'_M < p'_H; \\ 1, & \text{otherwise.} \end{cases} \tag{18}$$

That is, the attack probability is equal to 1 if and only if $p'_M \geq p'_H$. This condition is equivalent to the condition $p'_M \geq 1/2$. Using Lemma 1, we can rewrite this inequality as

$$1 - e^{-\alpha_M \Delta} p_H \geq \frac{1}{2},$$

that with an elementary transformations can be reduced to the inequality

$$2\big(1 - p_M\big) \leq e^{\alpha p_M \Delta}. \tag{19}$$

That is, the value p_0 is the minimal value of p_M, for which the inequality (19) holds, i.e., p_0 is the solution of the equation $2(1 - p_M) = e^{\alpha p_M \Delta}$, that we were to prove. □

Corollary 1. *Threshold decreases when parameter Δ increases (and, correspondingly, γ increases).*

Proof. According to (15), security threshold p_{st} is a function of Δ:

$$p_{st} = p_{st}(\Delta).$$

Show that this function is decreasing. After derivation of (15), we have:

$$-p'_{st} = \frac{1}{2}e^{\alpha \Delta p_{st}}(\alpha p_{st} + \alpha \Delta p_{st}),$$

from where we obtain

$$p'_{st} = -\frac{\alpha p_{st} e^{\alpha \Delta p_{st}}}{2 + \alpha e^{\alpha \Delta p_{st}}}.$$

Right part of the last equality is negative, so $p'_{st} < 0$ and p_{st} decreases when Δ increases. □

In Table 2 we give numerical results for the security threshold for various values of $\gamma = \gamma(\alpha, \Delta) = \alpha \cdot \Delta$.

Fig. 1 shows dependence of security threshold on the parameter γ.

E.g., for Bitcoin, $\alpha = 1/600$ and it is assumed that $\Delta = 20$ s (in practice Δ may take values from 1–2 s to 40 s [13]). For these

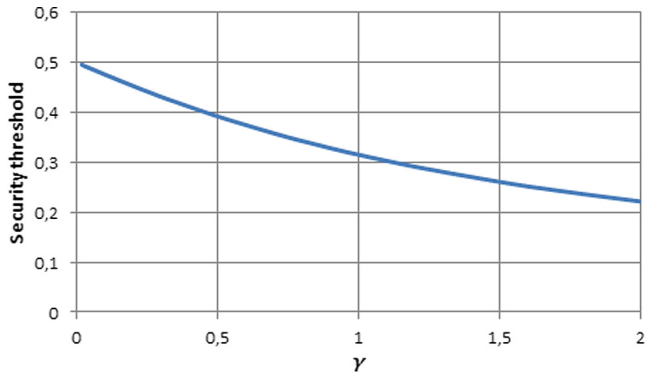


Fig. 1. Dependence of Security Threshold on the Parameter $\gamma = \Delta\alpha_H$ (the product of time delay difference and intensity of block generation).

Table 2
Security threshold for various values of parameter $\gamma = \gamma(\alpha, \Delta) = \alpha \cdot \Delta$.

γ	1/30	0.1	0.5	1	2
p_{st}	0.491737	0.475643	0.391798	0.314923	0.221427

parameters we obtain $\gamma = 1/30$ and the security threshold is $p_{st} = 0.491737$. It means that if the adversarial ratio is not less than 0.491737, his attack will be successful with probability 1.

According to Theorem 1, security threshold decreases when γ increases. As $\gamma = \Delta\alpha$, then we can also say that security threshold decreases when Δ or/and α increase. So for large enough parameter γ the ratio of adversary, which is enough for double spend attack with probability 1, may be much less than 50% (see Table 1). And vice versa: if $\Delta = 0$, then for any value of intensity α, not less than 50% ratio of adversary is enough to attack with probability 1. This consideration stress the significance of parameter Δ in analysis of network security.

5. Solution of the inverse problem: Finding upper bounds for intensity of block generation

Using Theorem 1, we also may solve another important problem: to determine the maximal intensity of block creation at which the network remains resistant to a double spend attack. Theorem 2 gives the solution of this problem.

Theorem 2. *For a given network with parameters p_H, p_M, Δ_H and Δ_M, the network is completely (with probability 1) vulnerable to a double spend attack if and only if the intensity α of block generation satisfies the following equality:*

$$\alpha \geq \frac{\ln 2p_H}{(1-p_H)\Delta} \tag{20}$$

(or $\alpha \geq \frac{\ln 2p_H}{p_M \Delta}$, which is the same).

In Table 3 we give the numerical results for maximal value of intensity of block generation (20), at which the network remains resistant to double spend attack, for different ratio of adversary.

E.g., for Bitcoin, if $\Delta = 20$ s and $p_M = 0.3$, the intensity may be increased by almost 33 times to almost 0.055 blocks per second. However, in this case the probability of unintentional fork will also increase, whereby a lot of work will be wasted.

6. Double spend attack probability with time delay

Now we find analytical expressions which connect, from the one hand, probability of double spend attack, and from the other hand — network parameters, such as synchronization time, intensity of block generation and hashrate of adversary. In what follows we make

Table 3
Maximal intensity α of block generation for various adversarial ratios and various Δ.

p_M	Δ				
	1 s	5 s	10 s	20 s	60 s
0.1	5.878	1.176	0.588	0.294	0.098
0.2	2.350	0.470	0.235	0.118	0.039
0.3	1.122	0.224	0.112	0.056	0.019
0.4	0.456	0.091	0.046	0.023	0.008
0.45	0.212	0.042	0.021	0.011	0.004

one more assumption in favor of adversary. We will assume that his synchronization time is equal to zero: $D_M = 0$. Note that analytical expression for attack probability, obtained under such assumption, may be also used in case when $D_M \neq 0$, but in this case the expression is the upper bound of the probability rather than its exact value.

For fixed $t \geq 0$ define the random variable $N_M(t)$ as the number of blocks which adversary creates and shares during time t.

Lemma 3. *Random variable $N_M(t)$ has Poisson distribution with parameter α_M:*

$$P(N_M(t) = n) = \frac{(\alpha_M t)^n e^{-\alpha_M t}}{n!}. \tag{21}$$

Proof. The event $\{N_M(t) = n\}$ may be represented as $\{S_M(n) < t\} \cap \{S_M(n+1) > t\}$, where random variable $S_M(n)$ was defined in (9). Then we may write the such chain of equalities:

$$\{N_M(t) = n\} = \{S_M(n) < t\} \cap \{S_M(n+1) > t\} =$$
$$= \{S_M(n) < t\} \cap \overline{\{S_M(n+1) < t\}} = \{S_M(n) < t\} \backslash \{S_M(n+1) < t\}.$$

But, according to (9), $\{S_M(n+1) < t\} \subset \{S_M(n) < t\}$, then, using (10), we obtain:

$$P(N_M(t) = n) = P\{S_M(n) < t\} - P\{S_M(n+1) < t\} =$$
$$= F_{S_M(n)}(t) - F_{S_M(n+1)}(t) = \frac{(\alpha_M t)^n e^{-\alpha_M t}}{n!},$$

and equality (21) is proved. □

Note 1. From the properties of Poisson processes (independent increments, absence of afteraction) for arbitrary $t_1, t_2 > 0$ we have:

$$N_M(t_1 + t_2) = N_M(t_1) + N_M(t_2). \tag{22}$$

This property will be used below in proof of the next lemma.

Define the random variable

$$X'_M(n) = N_M(S'_H(n)), \tag{23}$$

which is equal to the number of blocks created by adversary during the time when honest miners were created and shared n blocks.

Let us also define the random variable

$$X_M(n) = N_M(S_H(n)), \tag{24}$$

which is equal to the number of blocks created by adversary during the time when honest miners were created n blocks in case if there synchronization time is equal to zero.

Let build the distribution function of random variable $X'_M(n)$.

Lemma 4. *Define the probability $P_n(k)$ as*

$$P_n(k) = P(X'_M(n) = k). \tag{25}$$

Then the next statements hold.

1. *Random variable X'_M is a sum of two random variables:*

$$X'_M = X_M + N_M(nD_H) = N_M(S_H(n)) + N_M(nD_H), \tag{26}$$

where $S_H(n)$ and $X_M(n)$ were defined in (6) and (24), respectively.

2. *Random variable $N_M(S_H(n))$ has negative binomial distribution with parameters (n, p_H), and random variable $N_M(nD_H)$ has Poisson distribution with parameter $\alpha_M nD_H$:*

$$P(N_M(S_H(n))) = C^k_{n+k-1} p^n_H p^k_M, \tag{27}$$

$$P(N_M(nD) = k) = \frac{e^{-\alpha_M nD_H} \cdot (\alpha_M nD_H)^k}{k!}. \tag{28}$$

3. *The distribution of random variable $X'_M(n)$ is the next:*

$$P_n(k) = \frac{p^n_H}{(n-1)!} \cdot \frac{e^{-\alpha_M nD_H} \cdot (\alpha_M nD_H)^k}{k!} \times \times \sum_{i=0}^{k} \frac{(n-i+1)! \cdot C^i_k}{(\alpha nD_H)^i}, \tag{29}$$

where

$$\alpha = \alpha_M + \alpha_H.$$

Proof.

1. According to the definitions (23), (5), and (7), obtain:

$$X'_M(n) = N_M(S'_H(n)) = N_M(S_H(n) + nD_H) = = N_M(S_H(n)) + N_M(nD_H),$$

where the last equality is obtained according to (22), and equality (26) is proved.

2. To prove the equality (27) note that the random variable $N_M(S_H(n))$ is the same as the random variable X_n from the article [4], which, as was proved in this article (Proposition 5.1), has negative binomial distribution with parameters (n, p_H). Next, according to Lemma 3, $N_M(nD_H)$ has Poisson distribution with parameter αnD_H, from which we obtain (28).

3. According to the statement 1 of this Lemma,

$$N_M(S'_H(n)) = N_M(S_H(n)) + N_M(nD_H),$$

where the both items in the right part are independent random variables. So, according (25), the distribution function $P_n(k)$ of their sum is the convolution of their distributions:

$$P_n(k) = \sum_{i=0}^{k} \left(P(N_M(S_H(n)) = i) \cdot P(N_M(nD_H) = k - i) \right) = = \sum_{i=0}^{k} \left\{ C^i_{n+i-1} \cdot p^n_H \cdot p^i_M \cdot \frac{e^{-\alpha_M nD_H} \cdot (\alpha_M nD_H)^{(k-i)}}{(k-i)!} \right\} = = \frac{p^n_H}{(n-1)!} \cdot \frac{e^{-\alpha_M nD_H} \cdot (\alpha_M nD_H)^k}{k!} \cdot \sum_{i=0}^{k} \frac{(n-i+1)! \cdot C^i_k}{(\alpha nD_H)^i},$$

where $\alpha = \alpha_H + \alpha_M$. □

Now we are ready to prove the main result about the probability of double spend attack.

Theorem 3. *The probability $P(z)$ of double spend attack after z confirmation blocks, created by honest miners, is:*

$$P(z) = \begin{cases} 1, & \text{if } p'_M \geq p'_H; \\ 1 - \sum_{k=0}^{z} P_z(k) \left(1 - \left(\frac{p'_M}{p'_H} \right)^{z-k} \right), & \text{otherwise}, \end{cases} \tag{30}$$

where the random variable was defined in (29).

Proof. For the fixed z define the event $A_z(k)$ as

$$A_z(k) = \{ N_M(S'_H(z) = k) \} = \{ X'_M(z) = k \}, k \geq 0,$$

where the random variable $X'_M(z)$ was defined in (23).

Also define the event A_z as "adversary managed to realize double spend attack after z confirmation blocks, created by honest miners". Then

$$A_z = \left\{ \bigcup_{k \geq z} A_z(k) \right\} \cup \left\{ \bigcup_{k=0}^{z-1} (A_z(k) \cap E_{n-k}) \right\},$$

where the event E_{z-k} was defined in 2.

Note that the events $\left\{ \bigcup_{k \geq z} A_z(k) \right\}$ and $\left\{ \bigcup_{k=0}^{z-1} (A_z(k) \cap E_{n-k}) \right\}$ have empty intersection, and the events $A_z(k)$ and E_{z-k} are independent. Next, according to (18), $P(A_z(k)) = P_z(k)$, and according to Lemma 2, $P(E_{z-k}) = \left(\frac{p'_M}{p'_H} \right)^{z-k}$.

Using these two equalities we obtain:

$$P(A_z) = \sum_{k=z}^{\infty} P(A_z(k)) + \sum_{k=0}^{z-1} P(A_z(k)) \cdot P(E_{z-k}) = 1 - \sum_{k=0}^{z-1} P(A_z(k)) + \sum_{k=0}^{z-1} P(A_z(k)) \cdot P(E_{z-k}) = = 1 - \sum_{k=0}^{z-1} P(A_z(k))(1 - P(E_{z-k})) = 1 - \sum_{k=0}^{z-1} P_z(k) \left(1 - \left(\frac{p'_M}{p'_H} \right)^{z-k} \right),$$

and the equality (30) is proved. □

Using (30), we get the next two values:

- probability of double spend attack under defined network parameters (intensity of block generation, synchronization time, ratio of malicious miners) and fixed number of confirmation blocks;
- the minimal number of confirmation blocks (under defined parameters) that guarantees that the probability of double spend attack is less than given small value.

Tables 4 and 5 give the minimal number of confirmation blocks for which the probability of attack is not greater than 10^{-3}, for different network parameters.

Table 4
The minimal number z of confirmation blocks for which $p(z) \leq 10^{-3}$ for different network parameters Δ (in seconds) and p_M, and for $\alpha = 0.00167$ (as for BTC)

p_M	Δ			
	0 s	15 s	30 s	60 s
0.1	6	6	6	6
0.15	9	9	9	9
0.2	13	13	14	14
0.25	20	20	21	22
0.3	32	33	35	39
0.35	58	62	67	78
0.4	133	150	170	224

Table 5
The minimal number z of confirmation blocks for which $p(z) \leq 10^{-3}$ for different network parameters Δ (in seconds) and p_M, and for $\alpha = 0.0167$.

p_M	Δ			
	0 s	15 s	30 s	60 s
0.1	6	7	8	10
0.15	9	11	13	19
0.2	13	17	22	42
0.25	20	28	43	172
0.3	32	54	113	$P(z) = 1$
0.35	58	137	$P(z) = 1$	$P(z) = 1$
0.4	133	202	$P(z) = 1$	$P(z) = 1$

7. Conclusions

The paper shows how the intensity of block generation affects the network security, and it is presented accurate analytic expressions for the network security threshold, the upper bound of acceptable block generation rate, and the corresponding double spend attack probability. At the same time, it is essential that increase in the intensity of block generation results in making the network vulnerable to attacks, and, also the number of orphaned blocks is increased, i.e., the amount of wasted work is also increased.

Consequently, the problem of fast transaction processing, which becomes ever more important, cannot be solved in the classical blockchain approach with Nakamoto consensus. Therefore, more complex data structures should be used, like a DAG (Directed Acyclic Graph, [5,6,14,15]), Parallel Chains [16] or Prism [17] that significantly increase the block generation rate (and, accordingly, the speed of transaction processing) without compromising the security level. These protocols may be less vulnerable in case of large time delay due to possibility for several miners to confirm a block simultaneously.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Lyudmila Kovalchuk: Conceptualization, Methodology, Writing - original draft. **Dmytro Kaidalov:** Validation, Writing - review & editing. **Andrii Nastenko:** Software, Investigation. **Mariia Rodinko:** Writing - original draft, Data curation. **Oleksiy Shevtsov:** Visualization, Investigation. **Roman Oliynykov:** Supervision, Conceptualization, Validation, Writing - review & editing.

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